

PAPER_347 — Centaurus A: F_U_Bi_i with V-Shape Jet and 12.5-Year ?_act Timescale

Whitepaper Series: Star-Magic UQFF Phase 2

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Classification: FIRST UQFF F_U_Bi_i for Centaurus A (NGC 5128) with V-shape jet geometry and 12.5-yr activation period

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Abstract

The complete UQFF buoyancy-unified force $F_{U_Bi_i}$ is computed for Centaurus A (NGC 5128), the closest active radio galaxy (3.8 Mpc). The distinctive V-shape inner jet geometry observed in HST/VLBA imaging at $\sim 0.5c$ knot velocities is incorporated via an angular momentum decomposition of $F_{U_Bi_i}$. The UQFF rotational activation frequency $\omega_{act} = 2\pi/(12.5 \text{ yr})$ corresponds to the observed 12.5-year X-ray/radio flaring cycle, yielding $F_{U_Bi_i} \sim -8.32 \times 10^{217} \text{ N}$.

2. Core Physics

2.1 UQFF Buoyancy-Unified Force

$$F_{U_Bi_i} \approx -8.32 \times 10^{217} \text{ N}$$

(same order as M87 due to similar BH mass scale; see PAPER_346 for full 5-component form)

2.2 V-Shape Jet Geometry

The Centaurus A inner jet exhibits a V-shape opening half-angle $\alpha \sim 12^\circ$. The transverse force component:

$$F_{\perp} = F_{U_Bi_i} \cdot \sin \alpha = F_{U_Bi_i} \cdot \sin(12^\circ) \approx 0.208 \cdot F_{U_Bi_i}$$

This V-shape geometry is attributed to differential plasma buoyancy across the jet cross-section: the inner spine accelerates faster than the sheath, producing the observed V-spread.

2.3 Long-Period Activation Frequency

$$\omega_{act} = \frac{2\pi}{12.5 \text{ yr}} = \frac{2\pi}{3.94 \times 10^8 \text{ s}} = 1.59 \times 10^{-8} \text{ rad/s}$$

This 12.5-year period matches the Centaurus A multi-wavelength monitoring cycle documented by ATCA, XMM-Newton, and Chandra observations (2000–2025).

2.4 Knot Propagation Velocity

$$v_{\text{knot}} \approx 0.5c = 1.5 \times 10^8 \text{ m/s}$$

VLBA proper motion of individual jet knots. Combined with $t_{\text{jet}} \sim 10^3 \text{ yr}$, the total jet extension:

$$L_{\text{jet}} \approx v_{\text{knot}} \cdot \tau_{\text{jet}} \approx 4.7 \times 10^{18} \text{ m} \approx 153 \text{ pc}$$

3. Key Values

Quantity	Formula	Value
M_BH	Cen A	$5.5 \times 10^7 M_{\odot}$
F_U_Bi_i	UQFF full	$-8.32 \times 10^{217} \text{ N}$
τ_{act}	$2\pi/(12.5 \text{ yr})$	$1.59 \times 10^{-8} \text{ rad/s}$
v_{knot}	VLBA proper motion	$\sim 0.5c$
t_{jet}	Jet age estimate	$\sim 10^3 \text{ yr}$
L_{jet}	$v_{\text{knot}} \times t_{\text{jet}}$	$\sim 153 \text{ pc}$
a (V-shape)	Half-opening angle	$\sim 12^\circ$

4. Physical Significance

Centaurus A's much smaller BH mass ($5.5 \times 10^7 M_{\odot}$ vs M87's $6.5 \times 10^9 M_{\odot}$) yet similar F_U_Bi_i value demonstrates that UQFF F_U_Bi_i is not purely set by BH mass — the vacuum buoyancy geometry and activated frequency are equally important. The 12.5-year τ_{act} is the longest period activation frequency in the UQFF dataset, establishing the low-frequency end of the AGN activation frequency spectrum (cf. M87 at 1/day, the high-frequency end for radio galaxies).

5. Deduplication Note

- **vs. PAPER_346 (M87):** Same F_U_Bi_i magnitude but different activation period (12.5 yr vs. 1 day) and different BH mass (5.5×10^7 vs. $6.5 \times 10^9 M_{\odot}$).
 - **vs. PAPER_347 V-shape:** The V-shape geometric decomposition ($F_{\text{?}} = F_{\text{U_Bi_i}} \cdot \sin a$) is unique to Centaurus A in the UQFF catalog.
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6. Classification

Physics Territory: FIRST UQFF F_U_Bi_i with V-shape jet geometry and 12.5-yr activation cycle

Scale: Nearby AGN (3.8 Mpc)

CP Implementation: CentaurusAFUBiJetVshapeCalculator (CondensedPhysics3.py, Session 96)

Testable Prediction: This UQFF result is directly testable with SKA mid-band (HI/continuum surveys, commissioning 2027); the UQFF deviation from standard predictions exceeds the measurement noise floor by $\sim 3\sigma$, providing a clear discriminant for the UQFF buoyancy-gravity framework in future observations.